

Herding as the Contragredient of Conformal Prediction

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Abstract

The contragredient of a group action is the same action carried by the dual space: transform the functionals instead of the points. We show that this is the exact relation between conformal calibration, whose validity proof transports an evaluation functional over fixed scores, and herding, which transports points against fixed functionals; both control one argument of the bilinear pairing $\langle h, \nu \rangle = \int h d\nu$ while a cumulative discrepancy stays bounded by a constant, which is where both $O(1/n)$ rates come from. The relation is useful exactly insofar as proofs cross it, and we cross it in both directions through a stability lemma for the pairing: the coverage floor of Barber and Pananjady (2026) for split conformal on β -mixing series factors into the lemma’s two Lipschitz legs — buffer deletion as a measure perturbation, the switch coefficient as a distributional total variation — which is why their rate is linear in τ/n where blocking arguments give $\sqrt{\tau/n}$; and, in the reverse direction, the herding argument transfers to the calibration set, so that a low-discrepancy design makes realized coverage deterministic to $O(1/n)$. The conformal mechanism (late 1990s) predates herding (2009); the contragredient is herding.

1 Introduction

Two finite-sample constructions, developed independently, obtain $O(1/n)$ rates where sampling arguments give $O(n^{-1/2})$: the calibration step of conformal prediction, whose mechanism dates to the confidence machines of the late 1990s (see Vovk et al., 2005, for the canonical treatment, and Lei et al., 2018, for the split form used here), and herding (Welling, 2009).

The relation between them is named in the title, so we define the name at once. Given a group acting on a vector space, the *contragredient* is the induced action on the dual space: if points transform by $\rho(\pi)$, functionals transform by $\rho^*(\pi) = \rho(\pi^{-1})^\top$, and the pairing of a functional with a point is preserved. Herding acts on points: it selects $x_1, x_2, \dots$ so that fixed test functionals average correctly. Conformal calibration acts on functionals: the scores are never moved, the acceptance threshold is steered, and the validity proof transports the functional “evaluate the test coordinate” across the sample. One mechanism is the other with the action moved to the dual side.

A relation of this kind earns its keep only if arguments cross it. Sections 2 and 3 give the two mechanisms parallel formal statements; Section 4 states the duality on the pairing $\langle h, \nu \rangle = \int h d\nu$; Section 5 crosses it in both directions, factoring the dependent-data coverage floor of Barber and Pananjady (2026) through a stability lemma for the pairing and transferring the herding argument to the design of calibration sets. Remarks, including the relation to the Bayesian-quadrature reading of Snell and Griffiths (2025), close in Section 6.

2 Conformal calibration

Fix a miscoverage level $\alpha \in (0, 1)$ and write $p = 1 - \alpha$. Let $S_1, \dots, S_n$ be calibration scores and S_{n+1} a test score, assumed almost surely distinct, and let $k = \lceil p(n+1) \rceil$, assumed $\leq n$. Split conformal prediction declares coverage when $S_{n+1} \leq q_n$, where $q_n = S_{(k)}$ is the k -th smallest calibration score.

The choice of q_n is a feedback rule on the acceptance test $h_q(s) = \mathbf{1}\{s \leq q\}$.

Lemma 1 (Steering identity). *With $q_n = S_{(k)}$ and the scores distinct,*

$$\sum_{i=1}^n (h_{q_n}(S_i) - p) = k - pn \in [p, 1 + p).$$

In particular the cumulative acceptance-rate discrepancy is bounded by an absolute constant independent of n , and the empirical acceptance rate differs from p by at most $(1 + p)/n$.

Proof. Exactly k of the n calibration scores satisfy $S_i \leq S_{(k)}$, so the sum equals $k - pn$. Since $p(n+1) \leq k < p(n+1) + 1$, we have $p \leq k - pn < 1 + p$. $\square$

The control-theoretic reading of conformal prediction is established in the online setting: the level update of adaptive conformal inference (Gibbs and Candès, 2021) is integral feedback on realized miscoverage, and Angelopoulos et al. (2023) make the identification explicit as proportional–integral–derivative control. There the controlled quantity is the time-average miscoverage along a sequence; Lemma 1 is the batch counterpart, a single dead-beat correction applied to the calibration set before any test data arrive.

Validity rests on a deterministic counting identity concerning the *placement* of the test index.

Proposition 1 (Counting identity). *Let $s_1, \dots, s_{n+1}$ be distinct reals and, for $i \in \{1, \dots, n+1\}$, call placement i covered if s_i is at most the k -th smallest of $\{s_j\}_{j \neq i}$. Then placement i is covered if and only if the rank of s_i among all $n+1$ values is at most k , and consequently exactly k of the $n+1$ placements are covered.*

Proof. Placement i is covered iff at most $k-1$ of the other n values are smaller than s_i , iff the rank of s_i among all $n+1$ values is at most k . The ranks are a permutation of $\{1, \dots, n+1\}$, so exactly k indices qualify. $\square$

Corollary 1 (Marginal validity). *If $(S_1, \dots, S_{n+1})$ is exchangeable with almost surely distinct components, then $\mathbb{P}\{S_{n+1} \leq q_n\} = k/(n+1) \geq 1 - \alpha$.*

Proof. The coverage event $\{S_{n+1} \leq q_n\}$ is the event that placement $n+1$ is covered, since q_n is the k -th smallest of the values other than S_{n+1} . By exchangeability the indicator of “placement i is covered” has the same expectation for every i ; average Proposition 1 over i . $\square$

No probabilistic inequality enters before the final step, and the final step is an identity in distribution rather than a concentration bound.

3 Herding

Let $\phi : \mathcal{X} \rightarrow \mathcal{H}$ be a feature map into a Hilbert space and $\mu = \mathbb{E}_P \phi(X)$ a target mean. Herding (Welling, 2009) generates points by the recursion

$$x_t = \arg \max_{x \in \mathcal{X}} \langle w_{t-1}, \phi(x) \rangle, \quad w_t = w_{t-1} + \mu - \phi(x_t), \quad (1)$$

with w_0 given.

Proposition 2 (Herding discrepancy). *For any sequence produced by (1),*

$$\left\| \frac{1}{n} \sum_{t=1}^n \phi(x_t) - \mu \right\| = \frac{\|w_0 - w_n\|}{n}.$$

In particular, if $\sup_t \|w_t\| \leq B$ then the feature-mean discrepancy is at most $2B/n$.

Proof. Telescoping (1) gives $w_n = w_0 + n\mu - \sum_{t \leq n} \phi(x_t)$; rearrange and divide by n . $\square$

Conditions under which the weights remain bounded are given by Welling (2009) in finite dimension and analysed in general via the identification of (1) with the Frank–Wolfe algorithm on the marginal polytope (Bach et al., 2012); Chen et al. (2010) treat the kernel case. Under such conditions herding attains $O(1/n)$ where i.i.d. sampling attains $O(n^{-1/2})$.

4 The duality

Sections 2 and 3 exhibit the same ledger: a cumulative discrepancy — $k - pn$ in Lemma 1, $w_0 - w_n$ in Proposition 2 — bounded by a constant and divided by n . To compare the mechanisms, write both in terms of the bilinear pairing between bounded test functions h and finite measures ν ,

$$\langle h, \nu \rangle = \int h d\nu.$$

Each mechanism controls one argument of this pairing and holds the other fixed. Herding fixes the family of tests (the coordinates of ϕ , whose ν -averages are to match μ) and constructs the measure, $\nu_n = \frac{1}{n} \sum_t \delta_{x_t}$. Conformal calibration fixes the measure — the empirical distribution of scores — and constructs the test h_{q_n} ; its validity proof additionally moves the evaluation index, as follows.

Let ρ denote the permutation representation of the symmetric group $\mathfrak{S}_{n+1}$ on $\mathbb{R}^{n+1}$, $\rho(\pi)e_i = e_{\pi(i)}$, with contragredient $\rho^*(\pi) = \rho(\pi^{-1})^\top$, so that $\rho^*(\pi)e_j^* = e_{\pi(j)}^*$ and $\langle \rho^*(\pi)f, \rho(\pi)v \rangle = \langle f, v \rangle$.

Proposition 3 (Two descriptions of the orbit average). *For $s \in \mathbb{R}^{n+1}$ with distinct entries let $\chi_i(s) = \mathbf{1}\{s_i \leq Q_k(s_{-i})\}$, where $Q_k(s_{-i})$ is the k -th smallest entry of s with coordinate i deleted. Then for every $\pi \in \mathfrak{S}_{n+1}$,*

$$\chi_{n+1}(\rho(\pi)s) = \chi_{\pi^{-1}(n+1)}(s).$$

Consequently the average of χ_{n+1} over rearrangements of the data equals the average of $s \mapsto \chi_i(s)$ over placements i of the evaluation index — equivalently, over the orbit $\{\rho^(\pi)e_{n+1}^*\}$ of the evaluation functional at fixed data — and Corollary 1 may be proved on either side of the pairing.*

Proof. Q_k is invariant under permutations of its n arguments, and coordinate $n+1$ of $\rho(\pi)s$ is $s_{\pi^{-1}(n+1)}$; substitute. $\square$

Remark 1 (Direction, and what the name does). Permutation representations are orthogonal, so ρ^* is equivalent to ρ ; the contragredient adds no new representation, and its content here is the location of the action — on the covector side of the pairing for conformal, on the vector side for herding. Chronology fixes the direction of the phrase: the conformal mechanism predates herding by roughly a decade, so it is herding that realises the contragredient, in the primal space of generated points, with active selection of x_t replacing the transport of an evaluation index. The dependence analysis of Barber and Pananjady (2026) is more recent than either; the duality claims kinship of proofs, not priority of authors. Whether the relation is more than a name is settled by Section 5, where arguments cross it in both directions.

The correspondence is summarised below.

	Conformal calibration	Herding
controlled side	test h_{q_n} (and evaluation index)	measure ν_n
fixed side	empirical score measure	feature tests ϕ
ledger	$k - pn \in [p, 1 + p)$	$w_0 - w_n, \ w_t\ \leq B$
rate	$(1 + p)/n$	$2B/n$
probabilistic input	exchangeability	none (deterministic target)

Remark 2 (An image). In a gambling idiom: herding moves the balls to balance a fixed wheel; conformal prediction leaves the balls where they fell and relabels the slots under the pointer.

5 Moving arguments across the pairing

The vehicle is the joint Lipschitz property of the pairing.

Lemma 2 (Pairing stability). *For tests h, h' and finite measures ν, ν' ,*

$$|\langle h, \nu \rangle - \langle h', \nu' \rangle| \leq \|h - h'\|_\infty \nu'(\mathbf{1}) + \|h\|_\infty \|\nu - \nu'\|_{\text{TV}},$$

where $\|\cdot\|_{\text{TV}}$ is the total-variation norm. Perturbations of either argument cost linearly, and nothing concentrates.

Proof. Add and subtract $\langle h, \nu' \rangle$ and bound each difference by Hölder. $\square$

5.1 Toward conformal: the Barber–Pananjady floor

We first state the result exactly. For a score vector $S = (S_1, \dots, S_{n+1})$ and parameters k, τ , Barber and Pananjady (2026) define two subvectors of length $n + 1 - \tau$: $\Delta_{k,\tau}^0(S)$ deletes a block of τ entries while keeping S_{n+1} in the final slot, and $\Delta_{k,\tau}^1(S)$ deletes the same number while rotating S_k into the final slot. Their *switch coefficient* is the distributional distance

$$\Psi_{k,\tau}(S) = d_{\text{TV}}(\mathcal{L}(\Delta_{k,\tau}^0(S)), \mathcal{L}(\Delta_{k,\tau}^1(S))), \quad \bar{\Psi}_\tau = \frac{1}{n+1} \sum_{k=1}^{n+1} \Psi_{k,\tau}(S).$$

For a pretrained, memoryless score their Theorem 5 gives the floor

$$\mathbb{P}\{S_{n+1} \leq q_n\} \geq 1 - \alpha - \min_{0 \leq \tau \leq n} \left\{ \frac{\tau}{n+1} + \bar{\Psi}_\tau \right\}, \quad (2)$$

their Corollary 6 bounds $\bar{\Psi}_\tau \leq 2\beta(\tau)$ for stationary β -mixing sequences, their Theorem 8 supplies a matching ceiling, and their Theorem 7 shows (2) is tight over this class up to the factor $(1 - \alpha)/4$. Predictors with memory L and scores trained on an initial block are handled with the same structure at the cost of extra buffer terms.

The two steps of their argument instantiate the two legs of Lemma 2. The deletion step perturbs the measure argument: removing τ of $n + 1$ points moves the empirical score measure by at most $2\tau/(n + 1)$ in total-variation norm, and the image of that perturbation under the quantile functional is the level shift $\tau/(n + 1)$ in (2) — the insertion–deletion stability of order statistics is the pairing’s measure-side Lipschitz leg, specialised to h an indicator. The switch step perturbs on the test side in expectation: the evaluation functional is transported from slot $n + 1$ to slot k , and the price is,

by the definition of $\Psi_{k,\tau}$, a total variation between the laws of the two arrangements — the lemma applied to the distribution of the pairing rather than to a fixed realisation. Both legs are linear, so the floor is linear in τ/n ; a blocking argument (Oliveira et al., 2024) bounds the same quantity through an empirical-process maximal inequality and pays the Monte Carlo rate $\sqrt{\tau/n}$. Linearity is not an accident of their proof; it is the linearity of the pairing.

5.2 Toward herding: designed calibration

In the reverse direction the herding argument itself crosses the pairing: apply the bounded-discrepancy mechanism not to the generation of points for integration but to the selection of a calibration set.

Proposition 4 (Low-discrepancy calibration). *Let $\bar{F}$ be a continuous score distribution and suppose the calibration values $S_1, \dots, S_n$ (distinct, obtained by any design — herding, kernel thinning, a quantile grid) satisfy $\sup_s |\hat{F}_n(s) - \bar{F}(s)| \leq D$, where $\hat{F}_n$ is their empirical distribution function. Then the realized coverage of the split threshold against $\bar{F}$ satisfies, deterministically,*

$$|\bar{F}(S_{(k)}) - \frac{k}{n+1}| \leq D + \frac{1}{n}.$$

Proof. $\hat{F}_n(S_{(k)}) = k/n$, so $|\bar{F}(S_{(k)}) - k/n| \leq D$, and $k/n - k/(n+1) = k/(n(n+1)) \leq 1/n$. $\square$

A quantile grid achieves $D \leq 1/(2n)$; herding and kernel thinning achieve $D = O(1/n)$ up to logarithms in more general settings (Chen et al., 2010; Dwivedi and Mackey, 2021). Realized coverage is then within $O(1/n)$ of nominal on *every* draw: the dispersion of realized coverage collapses from the sampling order $n^{-1/2}$ to the herding order n^{-1} . The price is stated by the counting argument itself: selection by score value breaks the exchangeability of the calibration set with the test point, so the symmetry proof of Corollary 1 no longer applies and the guarantee is the deterministic one above, against the marginal law. The trade — symmetry for determinism — is the design of a measure-side controller for an index-side guarantee.

6 Remarks

Remark 3 (Quadrature). Snell and Griffiths (2025) recast the conformal computation as Bayesian quadrature over the calibration scores. That is a measure-side reading of the same pairing: quadrature weights on scores on one side, a transported evaluation functional on the other.

Remark 4 (Limits). The duality concerns the finite-sample ledger only. The passage from Proposition 1 to Corollary 1 is exchangeability’s, or, in Section 5.1, its mixing surrogate’s; no bookkeeping supplies it. Nor does control of thresholds and labels create information: it re-levels coverage while conditional sharpness remains a property of the scores (Cotton). Whether the steering can be aimed conditionally — selecting or reweighting calibration slots to balance the rank count given the input, as herding aims its particles at a target — is open; localized and weighted conformal methods are the nearest existing steps, and the same information-theoretic ceiling applies to them.

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